

Problem Set 2: Gains from Trade

Questions and Solutions

- 1) Aman is taking a 4th-year economics course. The class has a 2-person group project that needs a paper and a presentation. He can produce 1 page per hour or 5 mins of the presentation per hour, assuming he uses his time efficiently (i.e. assuming efficient production).
- 1a) Barbara can produce 2 pages per hour or 10 minutes of the presentation per hour, assuming efficient production. Who has the absolute advantage and who has the comparative advantage in paper writing?

Solution: Barbara can produce more pages per hour than Aman and more presentation minutes per hour, so **she has the absolute advantage in both goods.**

To find the comparative advantage, we need to work out who has the lower opportunity cost of paper writing.

If they're not writing the paper, they can work on the presentation. To produce one more page, Aman needs 1 hour, which means he gives up 5 presentation minutes.

To produce 1 extra page, Barbara needs 30 mins, and in those 30 mins she could have produced 5 presentation mins.

Her opportunity cost of 1 extra page is also 5 presentation minutes. **They have the same opportunity cost, so neither of them has a comparative advantage.**

Alternatively, you could calculate it using ratios. I recommend doing it this way: it catches the errors that are easy to make with intuitive arguments, especially when fractions are involved.

Aman's opportunity cost of paper writing is the number of presentation minutes he needs to give up for an additional page. We can calculate his presentation minutes per page using

$$\frac{\text{presentation minutes}}{\text{paper pages}} = \frac{\frac{\text{presentation mins}}{\text{hour}}}{\frac{\text{paper pages}}{\text{hour}}} = \frac{5}{1} = 5$$

That is, the (opportunity) cost of Aman writing an additional page is the 5 mins of presentation time he gives up.

Barbara's opportunity cost of paper writing is

$$\frac{\text{presentation minutes}}{\text{paper pages}} = \frac{\frac{\text{presentation mins}}{\text{hour}}}{\frac{\text{paper pages}}{\text{hour}}} = \frac{10}{2} = 5$$

Her (opportunity) cost of writing an additional page is also 5 mins of presentation time that she gives up.

- 1b) Suppose Aman and Barbara form a group. They each have a total of 10 hours to devote to this project. The paper needs to be 10 pages long. Who will write the paper and how long will their presentation be?

Solution: To be efficient, the person who starts writing the first page should be the one with the lowest opportunity cost, since they give up the least to produce that page.

In this case it doesn't matter: neither of them has a comparative advantage in paper writing, so **either one can write the paper, and the presentation will be 100 minutes long either way.**

If Aman spends 10 hours working on the paper, the total presentation time will be 0*5 minutes from him and 10*10 minutes from Barbara = 100 minutes. If Barbara writes the paper, producing 10 pages takes her 5 hours, which leaves 5*10 mins of presentation from her and 10*5 mins from Aman = 100 mins of presentation. If they share the paper writing, they still reach the same maximum number of presentation minutes.

- 1c) Barbara drops the course and Aman finds a new partner, Chen, who can produce 2 pages per hour or 12 mins of the presentation per hour, assuming efficient production. Who should write the 10-page paper now?

Solution: Chen also has an absolute advantage in both tasks. Her opportunity cost of 1 extra page is 6 mins of presentation, which is higher than Aman's, so **it is more efficient for Aman to write the paper.** He takes 10 hours to write it and gives up a total of 10*5 = 50 mins of presentation time to do it. If Chen wrote the paper, she would give up 10*6 mins = 60 mins to produce the same 10-page paper.

- 1d) The professor changes the requirement to 15 pages. On presentation day, Aman and Chen's team submitted their 15-page paper and did a 50-minute presentation. Were they on their PPF with this choice of paper length and presentation time? Explain.

Solution: As discussed above, if the team is producing efficiently, Aman should start the paper writing. After 10 hours he runs out of time, so Chen steps in to write the remaining 5 pages, which takes her 2.5 hours. With her remaining 7.5 hours she could produce a (7.5*12 mins =) 90-minute presentation, if the team were working efficiently. However, they only did a 50-minute presentation, which is less than their maximum possible, so **the team is below their PPF and producing inefficiently.**

- 1e) For the next project, Aman is assigned a different partner–Dov, who can produce 5 pages per hour. Since Dov is clearly better at paper writing, he should write the paper. Agree, Disagree or It depends? Explain.

Solution: It depends. Dov may have an absolute advantage in paper writing, **but until we know how good he is at presentation preparation, we can't say who should write the paper.** The parts above give examples that illustrate this.

- 2) Effie and Frank are roommates who have two household tasks to complete—cleaning and cooking. Effie takes 20 mins (i.e. $\frac{1}{3}$ of an hour) to clean a room and $\frac{3}{2}$ hours (i.e. 90 mins) to cook a meal, assuming efficient production. Frank takes 120 mins to clean a room and $\frac{5}{2}$ hours to cook a meal, assuming efficient production. They each have a total of 4 hours each to devote to household tasks over the weekend.

- 2a) What is Effie's opportunity cost of an additional clean room in terms of cooked meals and what is Frank's?

Solution: Please always assume that non-integer production is possible, unless explicitly stated otherwise. For example, here it is possible for Effie to produce 2.31 meals.

The difference between this question and the previous one is how productivities are expressed. In the previous question, you were given how many units (of paper or presentation) you could produce per hour. Here you are given how many hours (or mins) it takes to produce one unit. You can proceed in two ways.

(1) Convert all numbers into output produced per hour and then proceed as in the previous example. e.g. Effie produces 3 clean rooms per hour because each room takes $\frac{1}{3}$ hours. This may be better for consistency, so I've solved the question below this way.

(2) Figure out a conversion rate between clean rooms and meals using the amount of the resource it takes to produce each. For example, to get an additional clean room Effie needs 20 mins. Each meal takes 90 mins, so taking 20 mins away from cooking means giving up $\frac{20}{90} = \frac{2}{9}$ of a meal. This makes her opportunity cost of an extra clean room $\frac{2}{9}$ of a meal. I did this in minutes to show that you get the same answer in hours (below) or minutes, as long as you keep the units consistent.

For Effie, the number of rooms per hour is 3 and the number of meals per hour is $\frac{2}{3}$. To get an extra clean room, we need the conversion rate, meals per clean room:

$$\frac{\text{meals}}{\text{clean room}} = \frac{\frac{\text{meals}}{\text{hour}}}{\frac{\text{clean rooms}}{\text{hour}}} = \frac{\frac{2}{3}}{3} = \frac{2}{9}$$

so her opportunity cost of an extra clean room is $\frac{2}{9}$ meals.

To figure out how much an extra meal costs in terms of clean rooms, you can do the opposite. For an extra meal, the conversion rate between rooms and meals is

$$\frac{\text{clean rooms}}{\text{meal}} = \frac{\frac{\text{clean rooms}}{\text{hour}}}{\frac{\text{meals}}{\text{hour}}} = \frac{3}{\frac{2}{3}} = \frac{9}{2}$$

so her opportunity cost of an extra meal is $\frac{9}{2}$ clean rooms. Notice that it is the inverse of the opportunity cost of an extra room. You could do the same in minutes – rooms per minute and meals per minute – as long as the numerator and denominator use the same unit.

Now for Frank. He takes 120 mins (i.e. 2 hours) to clean a room, so he cleans $\frac{1}{2}$ rooms per hour, and he takes $\frac{5}{2}$ hours to cook a meal, so he cooks $\frac{2}{5}$ meals per hour. Running the same conversion for him,

$$\frac{\text{meals}}{\text{clean room}} = \frac{\frac{\text{meals}}{\text{hour}}}{\frac{\text{clean rooms}}{\text{hour}}} = \frac{\frac{2}{5}}{\frac{1}{2}} = \frac{4}{5}$$

so his opportunity cost of an extra clean room is $\frac{4}{5}$ of a meal.

You can check this in minutes, the way we did for Effie: a clean room costs Frank 120 mins and a meal costs him 150 mins, so a clean room costs him $\frac{120}{150} = \frac{4}{5}$ of a meal. Comparing the two, Effie gives up $\frac{2}{9}$ of a meal for an extra clean room and Frank gives up $\frac{4}{5}$, so Effie is the cheaper cleaner of the two.

- 2b) Suppose now that Frank moves out and Effie gets a new roommate, Faizal, who takes X hours to clean a room and $\frac{5}{2}$ hours to cook a meal, assuming efficient production. What value does X have to be for Effie to have an absolute advantage in cooking meals?

Solution: The value of X determines who has the absolute advantage in cleaning; it has nothing to do with cooking. In cooking, Effie has the absolute advantage no matter what X is, because she takes less time per meal and so produces more meals in an hour than Faizal.

- 2c) Over what range of X would you expect Effie to trade meals for Faizal's cleaning services?

Solution: In this case, Effie's trading with Faizal so we need to find the conditions under which she produces meals and exchanges them for cleaning services from Faizal.

If she cooks, it must be because Faizal produces cleaning services at a lower (opportunity) cost, and there are terms of trade that make them both better off.

Faizal's opportunity cost of cleaning in terms of meals is

$$\frac{\text{meals}}{\text{clean room}} = \frac{\frac{\text{meals}}{\text{hour}}}{\frac{\text{clean rooms}}{\text{hour}}} = \frac{\frac{2}{5}}{\frac{1}{X}} = \frac{2X}{5}$$

Which means we need

$$\frac{2X}{5} < \frac{2}{9} \Rightarrow X < \frac{5}{9}$$

If Faizal takes less than $\frac{5}{9}$ hours to clean a room, then Effie would cook and trade meals for cleaning.

- 3) Gerd and Helga are farmers who work for 10 hours a day. For each hour that Gerd works, he will end up with either 1.5 units of kale or 0.5 units of beef at the end of the season. Working for an hour, Helga can produce either 0.5 units of kale or 2 units of beef at the end of the season.

- 3a) If they trade, what is the minimum and maximum price of kale (in terms of beef) that will result?

Solution: We first calculate their opportunity costs of kale in terms of beef, to work out who produces which good for trade.

Gerd's opportunity cost of an additional unit of kale (hence it is in the denominator) in terms of units of beef (hence in the numerator) is

$$\frac{\text{units of beef}}{\text{units of kale}} = \frac{\frac{\text{units of beef}}{\text{hour}}}{\frac{\text{units of kale}}{\text{hour}}} = \frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{2} * \frac{2}{3} = \frac{1}{3} \text{ beef}$$

Helga's is

$$\frac{\text{units of beef}}{\text{units of kale}} = \frac{\frac{\text{units of beef}}{\text{hour}}}{\frac{\text{units of kale}}{\text{hour}}} = \frac{2}{\frac{1}{2}} = 2 * \frac{2}{1} = 4 \text{ beef}$$

Since Helga has a higher (opportunity) cost of kale, she will produce beef and trade it to Gerd for kale.

If she is going to buy Gerd's kale, the price has to be lower than what it costs her to produce it herself. So the highest price (i.e. the maximum price) for kale in this scenario is 4 units of beef. Anything more, and Helga is better off growing the kale she wants herself. Anything less, and she gains from the trade.

If Gerd is going to produce the kale, the lowest price he will accept is $\frac{1}{3}$ units of beef (i.e. the minimum price). Anything less, and he would rather move time out of kale production and raise the beef he wants himself. Anything higher, and he gains from the trade.

If they trade, the price of kale will lie between these two, i.e. between $\frac{1}{3}$ and 4 units of beef.

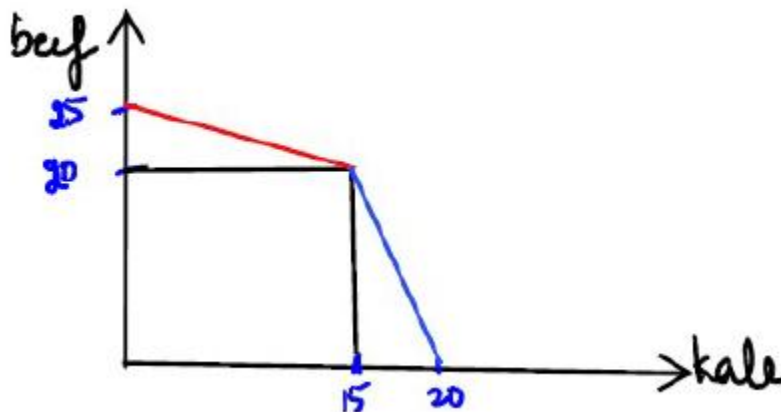


Figure 1: The joint PPF when Gerd (red) and Helga (blue) produce efficiently.

- 3b) Suppose that, after trade, the two of them are producing 21 units of beef between them. What is the opportunity cost of beef now?

Solution: We know that Helga is the producer with the lowest opportunity cost of beef. Why? Because the opportunity cost of beef is just the inverse of the opportunity cost of kale. For Helga it is

$$\frac{\text{units of kale}}{\text{units of beef}} = \frac{\frac{\text{units of kale}}{\text{hour}}}{\frac{\text{units of beef}}{\text{hour}}} = \frac{\frac{1}{2}}{2} = \frac{1}{4} \text{ kale}$$

and for Gerd it is 3 kale.

If we didn't know how much beef and kale were being produced after trade, the most we could say is that the opportunity cost of beef is either $\frac{1}{4}$ or 3 units of kale, depending on who is producing it. But we are told how much beef is being produced after trade, so we can say more.

If Helga spends all her time on beef production, she produces 10 hours * 2 units = 20 units of beef. Any additional beef has to come from Gerd, whose opportunity cost is the higher one of 3 units of kale.

If after trade they are producing 21 units of beef, then Gerd is producing 1 unit of beef, which takes him 2 hours, and with the remaining 8 hours he produces $8 * \frac{3}{2} = 12$ units of kale.

At 21 units of beef, the opportunity cost of beef is the cost faced by the producer who can still move resources into beef production. That person is Gerd, and **an extra unit of beef costs him (and society) 3 units of kale.**

- 3c) A consultant looking at the opportunity cost numbers says that Helga should produce only beef and Gerd only kale. Agree, Disagree or It depends?

Solution: It depends (a typical economist answer, though not always). It depends on the demand for beef (or kale).

If the demand for beef is very high, once the low (opportunity) cost producer, Helga, has produced all the beef she can, the extra beef to satisfy the demand will come from the high-cost producer, Gerd. **Since Gerd is also the lowest-cost producer of kale, he ends up producing both goods rather than specializing.**

To convince yourself, look at the numerical example in part (b).

- 4) You are really worried about a [zombie apocalypse](#) and have just bought a plot of land in rural Ontario. Your plan is to escape there if zombies attack. You are in college studying Economics, and you want to become a professor. However, that is not a very useful skill when society collapses. You have a choice right now: study to achieve your dream of becoming a professor, or learn skills that will help you survive when the zombies arrive. From a comparative advantage perspective, what would you choose? What does it depend on? For example, does it matter if you're taking friends with you to your safe house? Does it matter what skills your friends have?

Real world inspiration: [Preparedness 101: Zombie Apocalypse](https://blogs.cdc.gov/publichealthmatters/2011/05/101-zombie-apocalypse/) on the *Public Health Matters* blog of the Centers for Disease Control and Prevention. Posted May 16, 2011 available at: <https://blogs.cdc.gov/publichealthmatters/2011/05/101-zombie-apocalypse/>.

Solution: This question asks you to think about the role specialization plays in our lives, and how much that role depends on circumstances and on what is actually possible.

Choosing a career is choosing to specialize.

Hopefully, you specialize in something you have a comparative advantage in – that is, something you are relatively good at, given what everyone else in society can do. You can become a professor because you know that if you ever need your plumbing fixed, you can trade with someone who has a comparative advantage in that skill. (You could barter services, but money serves as a medium of exchange.)

Whether you become a professor could depend on how confident you are that you will find a plumber when you need one, and that the terms of trade will make it worth your while to hire them. If not, you may be better off doing it yourself.

What makes you so sure that you will find a plumber and pay them rather than learn to do it yourself? What are you implicitly assuming?

From society's perspective it is more efficient to allocate people to tasks based on comparative advantage (i.e. to be on the PPF), but that may not be possible: it depends on the extent of the market, people's expectations, and larger institutions.

- 5) The topic of Canadian jobs moving abroad is often in the news. The types of jobs and the industries change. One example is call-center jobs moving to India. Recently, those jobs have been moving back to Canada, as discussed in a *Globe and Mail* article. A CEO in the article explains why his company moved jobs back:

“It worked well – for a while,” Mr. Day said about Primus’s Indian experience, which involved the creation of new jobs to serve the company’s expanding consumer business in the mid-2000s. ‘We got qualified, college-educated employees with accent neutralization training. But over the years we felt the quality of agent started to decline’ and by 2009, ‘my money wasn’t buying what it was a few years earlier.’”¹

How could you explain this declining worker quality using the concepts we have studied in this topic?

Note: there could be many other reasons behind this change. Asking you to look at it through a single lens is constraining, and may force you to ignore important aspects. The purpose is to develop your ability to apply course concepts to a particular real-world example. Notice which important aspects feel missing when you focus on just one concept, and pay attention to why you think it is not the best concept to apply here. Doing this develops the other course learning objective—appreciating the role of assumptions and the benefits/limitations of models. Your answer may be different from mine, so discuss your reasoning with your classmates and us. Just use topic concepts.

Solution: One way to look at it is the beef-and-kale example from Q3b, applied within a single country. The mechanism there was this: once Helga, the low-cost producer, had produced all the beef she could, any extra beef had to come from Gerd, the high-cost producer, so the opportunity cost of the marginal unit of beef rose. Here, call-center services play the role of beef, and workers within India play the roles of Helga and Gerd.

Suppose India has a comparative advantage in call-center services relative to Canada and other richer countries e.g. the US and the UK. If demand for call-center services from India is high, the workers best suited to the job – the ones with the lowest opportunity cost, the Helgas – select into those jobs first. Once all the Helgas are employed and demand is still high, the Gerds start working at call-centers, and they are not as good at it. Given the Gerds' opportunity costs compared to those of Canadian workers, it may now make more sense to bring some jobs back.

Throughout this reasoning, Canadian workers could have had the absolute advantage in call-center services all along. But comparative advantage – and hence trade – depends on relative productivities, i.e. opportunity costs. You can use the numerical examples in the other problem set questions to think through this as well, e.g. the Effie/Frank/Faizal question.

¹ “Canadian call centres ring up new job growth” by Sean Silcoff in *The Globe and Mail*. Published March 13, 2012, updated May 2, 2018. Available at: <https://www.theglobeandmail.com/report-on-business/economy/jobs/canadian-call-centres-ring-up-new-job-growth/article533634/>

Checked with Claude (Anthropic) for grammar, clarity and errors. Tell me if you find one I missed.